

William M. Mongan, Jr., Ph.D.

CONTACT INFORMATION

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FACULTY APPOINTMENTS

Co-Director

Teaching and Learning Institute (TLI)
Collegeville, Pennsylvania

Ursinus College

2025-Present

Associate Professor

Mathematics and Computer Science
Collegeville, Pennsylvania

Ursinus College

2020-Present

- Tenure awarded: July, 2025
- Faculty Coordinator, Artificial Intelligence Minor, 2025-Present
- Associate Professor, 2022-Present
- Assistant Professor (Visiting), 2020-2022

Instructor

Computer Science and Engineering
Syracuse, New York

Syracuse University

2020-2020

- Instructor (Part-Time / Visiting), 2020-2020

Visiting Research Scientist
College of Engineering
School of Education
Philadelphia, Pennsylvania

Drexel University

2019-Present

- Collaborated on research projects related to mathematics teaching and learning using AI with the Drexel School of Education and Mathforum
- Co-Advised research students on funded research projects in signal processing and machine learning on biomedical smart textiles

Associate Department Head for Undergraduate Affairs

Computer Science
Philadelphia, Pennsylvania

Drexel University

2014-2019

- Associate Department Head for Undergraduate Affairs, 2015-2019
- Director of Undergraduate Affairs, 2014-2015

Lecturer

Computer and Information Science
School of Engineering and Applied Science
Philadelphia, Pennsylvania

University of Pennsylvania

2012-2013

- Lecturer (Part-Time / Visiting), 2012-2013

Teaching Professor

Computer Science
Electrical and Computer Engineering
Philadelphia, Pennsylvania

Drexel University

2008-2019

- Teaching Professor (full), 2017-2019
- Associate Teaching Professor, 2012-2017
- Assistant Teaching Professor, 2011-2012
- Instructor / Auxiliary Professor, 2008-2011

ACADEMIC
CONSULTING AND
BOARD
APPOINTMENTS

Advisory Board Member

Jumpstart Philadelphia CS for All
Researcher-Practitioner Partnership (RPP)
Advisory Board
2025-Present

Advisory Council Member
Montgomery County, PA

Artificial Intelligence Council for the Public Good
2025-Present

- Member of the Economic and Workforce Development Subcommittee

Member of the Board of Managers

Jim Fisher Memorial Digital Network
Association (JFMDNA)
2024-Present

Montgomery County, PA

- Served as Collegeville Borough ARES/RACES operator for Limerick Generating Station Drill with FEMA and PEMA, November, 2025
- Established AREDN Mesh Network DMR System for Montgomery County ARES/RACES, 2025
- Configured PBX system for Mesh Network portable demonstrations, 2025

Board Member, ITAG

Innovative Technology Action Group (ITAG),
Chester County Economic Development Council
(CCEDC)
2024-Present

Chester County, PA

- Organizing Committee Member for TECH360 Conference, 2025-Present

Vice President

Computer Science Teachers Association (CSTA),
Longwood Chapter
2022-Present

Berks-Chester-Lancaster-York Counties, PA

Educational Consultant
Newark, DE

Delaware Department of Education
2020-2020

- Developed workshop on teaching the AP Computer Science A course with a team of teachers
- Presented the AP Computer Science A workshop at Partner4CS at the University of Delaware in September, 2020

Educational Consultant
Pennsylvania Training and Technical Assistance
Network (PATTAN)
Harrisburg, PA

Pennsylvania Department of Education
2020-Present

- Member of the CS State Strategy Committee and of the CS Leadership and Data Subcommittee, 2025-2026
- Data co-lead for ECEP Common Metrics Project and code.org Statewide CS Data Collection and Reporting, 2025-Present
- co-Author of the CSinPA Landscape Report¹ for Computing Education in Pennsylvania, September, 2024.
- CSinPA 2024 Summit Instructional Planning Breakout Facilitator for “Determining Acceptable Evidence” and “Designing Lesson Activities,” December, 2024
- Member and Co-Chair of the Expanding Computer Science in Pennsylvania (CSinPA) Strategic Planning Committee under the Expanding Computing Education Pathways (ECEP) Alliance, Higher Education Workgroup, 2024-Present
- Virtual Community Leader on “Design Thinking as a Bridge from K-12 to the Workforce,” March-April, 2023
- Developed Grades 7-12 curriculum for Computer Science teacher preparation under the new Computer Science state teaching certification under a PASmart grant from the Pennsylvania Department of Education in partnership with the Lancaster-Lebanon Intermediate Unit

¹<https://drive.google.com/file/d/1qUiVW0yDdALVyAKehub5tqAE6nSCyPdt/view>

- Hosted Remake Learning Days Monthly Meetup as a Panel Moderator in May, 2022
- Invited Panel Moderator at CS4AllPA Summit in June, 2022
- Developed educational materials on computing career pathways and try-at-home activities
- Producer and Host of the [digitalsignature.fm](https://www.digitalsignature.fm/)² podcast with support from the Pennsylvania Training and Technical Assistance Network (PaTTAN) and the Pennsylvania Department of Education, June, 2022-Present

EDUCATION

Doctor of Philosophy

Drexel University

PhD in Electrical and Computer Engineering
Philadelphia, Pennsylvania

2013-2018

- PhD Dissertation³: *Predictive Analytics on Real-Time Biofeedback for Actionable Classification of Activity State*
- PhD Advisor: Dr. Adam Fontecchio
- Cumulative Graduate GPA: 3.91 on a 4.00 index

Selected Graduate Computer Science and Engineering Course List

- CS521 and CS522: Data Structures and Algorithms I and II
- CS575: Software Design
- CS550: Programming Languages
- CS680: Advanced Topics in Software Engineering: Cyber Forensics
- CS680: Distributed Systems
- CS543: Operating Systems
- CS680: Modeling for Snake Based Robots
- CS544: Networking
- CS525: Theory of Computation
- CS676: Parallel Programming
- CS680: Advanced Topics in Software Engineering: Modeling and Architecture
- CS576: Dependable Software Systems
- CS898: Master's Thesis
- ECE997/ECE997: Dissertation Research
- ECE998: PhD Dissertation

Master of Science

Drexel University

MS in Science of Instruction, School of Education
PA Instructional Certificate in Mathematics
Philadelphia, Pennsylvania

2005-2008

- Earned a Secondary Mathematics (Pennsylvania In-State: Mathematics Grades 7-12) teaching certification in Pennsylvania in July 2007, with a Computer Science subject area add-on in July, 2021
- GPA: 3.86 on a 4.00 index
- Served as a graduate fellow for the STEM educational outreach program in the NSF GK-12 (NSF Award DGE-0538476) Program at the School District of Philadelphia at two schools and grades 5-8 (2006-2008)
- Student taught grade 8 mathematics at Middle Years Alternative School in the School District of Philadelphia

Selected Graduate Teacher Education Course List

- EDUC520: Professional Studies in Instruction
- EDUC522: Evaluation of Instruction
- EDUC524: Current Research in Curriculum and Instruction
- EDUC525: Multimedia in Instructional Design
- EDUC528: Cultural and Historical Significance of Mathematics
- EDUC523: Diagnostic Teaching
- EDLT536: Learning Sciences and Instructional Design

²<https://www.digitalsignature.fm/>

³<https://www.billmongan.com/publication/dissertation>

- EDUC533: Virtual Learning Communities
- EDUC540: Field Experience (Student Teaching)
- EDUC775: Authentic Assessments

Master of Science

Drexel University

MS in Computer Science, College of Engineering
Philadelphia, Pennsylvania

2005-2008

- MS Thesis in Computer Science⁴: *A Service-Based Web Portal for Integrated Reverse Engineering and Program Comprehension*
- MS Thesis Advisor: Dr. Spiros Mancoridis
- Graduate Research Assistant for the College of Engineering under the Applied Communications and Information Networking (ACIN) Project, and for the Department of Computer Science under the Software Engineering Research Group (SERG)
- GPA: 3.86 on a 4.00 index

Bachelor of Science, Magna Cum Laude

Drexel University

BS in Computer Science, College of Engineering
Philadelphia, Pennsylvania

2000-2005

- Undergraduate Research Assistant
- GPA: 3.79 on a 4.00 index
- Minor in Mathematics, College of Arts and Sciences
- Teaching Assistant for Systems Architecture I and II (CS281, CS282): 2003-2005

Undergraduate Track Elective Concentrations

- Operating Systems
- Compiler Theory
- Software Engineering

TEACHING AND SCHOLARLY DEVELOPMENT

COURSES TAUGHT

Selected course web pages are available at: <https://www.billmongan.com/teaching>

- **Ursinus College**
 - Instructor and Author for Artificial Intelligence (CS477)
 - Co-Instructor for Pedagogical Partnerships (IDS060)
 - Instructor and Author for Foundations of Artificial Intelligence (CS357)
 - Guest Lecturer in Physics on Screen (PHYS110)
 - Guest Lecturer in Computer Networks (CS475)
 - Instructor and Author for AI for All (IDS099 Special Topics)
 - Instructor and Author for Operating Systems (CS376)
 - Instructor and Author for Problem Solving in CS (CS010)
 - Instructor and Author for Hidden Curriculum in CS (IDS099 Special Topics)
 - Instructor and Author for Computer Science Pedagogy (CS471 Special Topics, CS392)
 - Instructor for Common Intellectual Experience I (CIE100)
 - Instructor and Author for Computer Architecture (CS274)
 - Instructor and Author for Software Engineering (CS375)
 - Instructor for Advising Practicum (IDS-ADV)
 - Instructor and Author for Programming for the World Around Us (CS170Q)
 - Instructor and Author for Human-Computer Interaction (CS474)
 - Instructor and Author for Database Design (CS377)
 - Instructor and Author for Principles of Programming Languages (CS374)
 - Instructor and Author for Computer Networks (CS475)
 - Instructor and co-Author for Introduction to Computer Science (CS173)
 - Instructor and Author for Web and Mobile Development (CS471 Special Topics)
- **Syracuse University**

⁴<https://www.billmongan.com/publication/msthesis>

- Instructor and Course Coordinator for Advanced Data Structures and Algorithms (CSE674 Graduate Course)
- **University of Pennsylvania**
 - Instructor for Computer Systems II: Digital Systems Organization and Design (CIT595 Graduate Course)
- **Drexel University**
 - Instructor for Advanced Programming Tools and Techniques (CS265)
 - Instructor and Author for Programming Practicum (CS T280 Special Topics)
 - Instructor for Computer Science Foundations (CS520 Graduate Course)
 - Guest Lecturer for Collaborative Intelligent Radio Networks (CIRN) Design (ECE T680 Special Topics Graduate Course)
 - Instructor and co-Author for Computing and Informatics Design Project (CI106, CS280 Special Topics)
 - Instructor and co-Author for Computing and Informatics Design I (CI101)
 - Instructor and Author for “Making Apps” Honors Colloquium course for the university (HNRS202)
 - Instructor and Author for Processor Architecture and Analysis (CS352, CS480 Special Topics)
 - Instructor and Author for Web and Mobile Application Development (CS275, CS280, CS480 Special Topics, CS690 Special Topics Graduate Course)
 - Instructor for ESL097: the King Abdullah University of Science and Technology Introduction to C++ course for high school seniors for the Drexel University English Language Center (joint grant between Drexel University and the University of Pennsylvania)
 - Guest Lecturer for Operating Systems (CS543 Graduate Course)
 - Guest Lecturer for Computer Networks (CS544 Graduate Course)
 - Instructor and Guest Lecturer for Systems Architecture I and II (CS281, CS282)
 - Instructor for Computer Organization (ECEC490 Special Topics)
 - Instructor and Author for Systems Programming (CS283, ECEC353)
 - Instructor for Concurrent Programming (CS361, ECEC490 Special Topics)
 - Instructor for Processor Design (ECEC490 Special Topics)
 - Instructor and Author for Operating Systems (CS370, ECEC421, ECEC490 Special Topics)
 - Instructor for Computer Networks (CS472, ECEC357, ECEC490 Special Topics)
 - Instructor for Computer Structures (ECEC355)
 - Instructor and Author for Machine Organization (CS680 Special Topics Graduate Course)
 - Instructor for Internet Arch and Protocols (ECEC432)
 - Instructor for Network Programming (ECEC433)
 - Instructor and Author for Special Topics in Network Security (ECEC490 Special Topics)
 - Instructor for Programming for Engineers (ECE203)

INDEPENDENT STUDY AND INTERNSHIP SUPERVISION

- **Ursinus College**
 - Internship Advisor (CS382): 2 students, 2026-2027
 - Internship Advisor (CS381): 1 student, 2026-2027
 - Independent Study Advisor: Adaptive Prompt Evaluation and Refinement Loop (1 student, CS392)
 - Internship Advisor (CS382): 8 Students, 2025-2026
 - Independent Study Advisor: Independent Study in Digital Interaction in Performance Art (1 student, CS394)
 - Independent Study Advisor: Independent Study in Educational Pathways in Computing (1 student, CS394)
 - Independent Study Advisor: Independent Study in Software Engineering in CS (2 students, CS394)
 - Independent Study Advisor: Independent Study in Using Lightweight Machine Learning to Assess Risk in Network Port Configurations (1 student, CS392)
 - Independent Study Advisor: Independent Study in Cybersecurity and Telecommunications (1 student, CS392)

- Independent Study Advisor: Independent Study in RF Emergency Communications (1 student, CS394)
- Independent Study Advisor: Independent Study in Biomedical Artificial Intelligence (1 student, CS394)
- Independent Study Advisor: Honors Independent Study in AI in Teaching and Learning (1 student, CS491-CS492)
- Independent Study Advisor: Independent Study in AI in Teaching and Learning (3 students, CS394)
- Independent Study Advisor: Honors Independent Study in AI Based Wi-Fi Penetration Tool (1 student, CS491-CS492)
- Independent Study Advisor: Independent Study in Reliable Wireless Emergency Communications (1 student, CS394)
- Internship Advisor (CS382): 4 Students, 2024-2025
- Independent Study Advisor: Independent Study in FinLitKit: Financial Literacy Toolkit Development (CS392)
- Independent Study Advisor: Research Independent Study in Sensor Fusion (2 students, CS392)
- Independent Study Advisor: Research Independent Study in AI Feature Visualization (CS392)
- Independent Study Advisor: Research Independent Study in Natural Language Processing (3 students, CS394)
- Independent Study Advisor: Research Independent Study in Retrieval Augmented AI (CS394)
- Independent Study Advisor: Research Independent Study in AI Feature Visualization (CS391)
- Independent Study Advisor: Research Independent Study in Virtual Museum Environments (CS391)
- Independent Study Advisor: Research Independent Study in Advanced Computer Architecture and Organization (CS394)
- Independent Study Advisor: Research Independent Study in Human Pose Estimation through WiFi (2 students, CS394)
- Independent Study Advisor: Research Independent Study in RFID Localization and Tracking (3 students, CS394)
- Independent Study Advisor: Research Independent Study in RFID Dynamic Radar Tracking (CS394)
- Independent Study Advisor: Research Independent Study in Educational Modules for Grid Software Defined Radios (CS394)
- Independent Study Advisor: Research Independent Study in Grid Software Defined Radios (CS392)
- Independent Study Advisor: Research Independent Study in Explainable AI Models (CS392)
- Independent Study Advisor: Research Independent Study in RF IoT Security Layer (CS394)
- Independent Study Advisor: Research Independent Study in UX Design for Student Engagement (CS394)
- Independent Study Advisor: Research Independent Study in Moon Crater Detection (CS394)
- Independent Study Advisor: Research Independent Study in Moon Crater Detection (2 students, CS391)
- Independent Study Advisor: Research Independent Study in Neural Network Visualization (CS394)
- Independent Study Advisor: Research Independent Study in Malware Detection through Electromagnetic System Response (CS391)
- Independent Study Advisor: Research Independent Study in RF Biomedical Analysis through Wearable Internet-of-Things Devices (2 students, CS394)
- Independent Study Advisor: Research Independent Study in Moon Crater Classification (CS391)
- Independent Study Advisor: Research Independent Study in RF Biomedical Analysis through Wearable Internet-of-Things Devices (2 students, CS394)
- Independent Study Advisor: Research Independent Study in Moon Crater Classification (2

- students, CS392)
- Independent Study Advisor: Research Independent Study in Cross-Architecture Translation (CS392)
- Independent Study Advisor: Research Independent Study in IoT Wearable Biosensors (CS391)
- Independent Study Advisor: Research Independent Study in Dynamic Antenna Selection for Real-Time RF Sensing (CS391)
- Independent Study Advisor: Research Independent Study in Cross-Architecture Translation (CS394)
- Independent Study Advisor: Research Independent Study in Moon Crater Classification (CS391)
- Independent Study Advisor: Research Independent Study in Deep Learning in Astrophysics (CS391)
- **Drexel University**
 - Independent Study Advisor: RF Cardio RSA Monitoring (CS I599 Graduate Course)
 - Independent Study Advisor: Machine Learning Research (CS I399)
 - Independent Study Advisor: Cardio-respiratory RF Processing (CS I299)
 - Independent Study Advisor: Advanced Computing Systems (CS I499)
 - Independent Study Advisor: Coordinated Radio Frequency Management (HNRS I399)
 - Independent Study Advisor: Advanced Operating Systems (CS I499)
 - Independent Study Advisor: RF Multisensor Fusion (CS I499)
 - Independent Study Advisor: ECE Magnetic Knitting Investigation (CS I799 Graduate Course)
 - Independent Study Advisor: RF Biomedical Sensing (CS I299)
 - Independent Study Advisor: RFID Multisensor Fusion (CS I499)
 - Independent Study Advisor: Forecasting on Medical Data (CS I499, CS I199)
 - Independent Study Advisor: Biomedical Signal Processing (CS I499)
 - Independent Study Advisor: Advanced Web Services and Mobile Application Development (CS I499)
 - Independent Study co-Advisor: Social Media Analytics (CS498)
 - Independent Study Advisor: Computing and Informatics Design Experience (CS498)
 - Independent Study Advisor: Big Data and Advanced Analytics (CS498)
 - Independent Study Advisor: Advanced Topics in Operating Systems (CS498)
 - Independent Study Advisor: Special Topics in Computer Architecture (CS498)
 - Independent Study Advisor: Advanced Topics in CPU Design (CS498, ECEC497)
 - Independent Study Advisor: Systems Architecture I (CS498)
 - Independent Study Advisor: Systems Architecture II (CS498)
 - Independent Study Advisor: Concurrent Programming (CS498)

RESEARCH SUPERVISION

- **Ursinus College**
 - Faculty supervisor to one undergraduate student on AI-enabled analysis of faculty early alert tonality on student engagement and success outcomes, 2026
 - Faculty supervisor to one undergraduate and one graduate co-op student on AI research and development in mathematics teaching and learning, 2025-2026
 - Faculty supervisor to two Gates Foundation funded students on AI in mathematics teaching and learning, 2025-2026
 - Faculty Advisor to an undergraduate Andrews Fellow and Summer Fellow in Localization and Monitoring using Passive RF on the Internet-of-Things, 2023, who went on to be named a 2024 Goldwater Scholar
 - Faculty Advisor to two undergraduate students in localization using outdoor wireless access points, 2020-2021
 - Faculty Advisor to an undergraduate student in RESTful RFID interrogators for biomedical sensing on the Internet-of-Things, 2020-2021
 - Founder and Principal Investigator of the Human-Machine Intelligent Systems Lab⁵, 2020

⁵<https://www.hmislab.org/>

- **Drexel University**

- Co-Advisor to ECE Senior Design team in closed loop sensing and actuating systems for neonatal respiratory support, 2022-2023, 2023-2024
- Co-Advisor to ECE Senior Design team in IoT for Smart Campus Communities (VarIoT), 2022-2023; recognized for the best department senior project, 2023-2024
- Co-Advisor to ECE Senior Design team in closed loop sensing and actuating systems for neonatal respiratory support, 2021-2022
- Co-Advisor to ECE Senior Design team in IoT for Smart Campus Communities (VarIoT), 2021-2022
- Faculty co-Advisor for student group in IEEE Computer Society Competition (COMPSOC) project: *RFID and BLE Enabled Smart Textile Wearable for COVID-19 Response*, 2020-2021; project received honorable mention and team won the IEEE Philadelphia Section Merrill Buckley Jr. Student Project Award
- Faculty co-Advisor for senior design group in DVT symptom detection and mitigation, 2020-2021; group selected to represent the ECE Department at the College of Engineering Senior Design Competition and received third place at the College Competition
- Faculty Mentor to 4 Vertically Integrated Projects (VIP) students in RF-based indoor localization, 2019-2020
- Faculty Mentor to 3 VIP students in RF denoising through multisensor fusion with passive wearable IoT devices, 2019-2020
- Faculty Mentor to 5 VIP students in adaptive respiratory event prediction using passive wearable RF-based smart garment devices, 2019-2021
- Faculty co-Advisor to VIP student project in RF modulation classification, 2019-2020
- External Stakeholder for senior design group project: VarIoT software framework for the Internet of Things, 2019-2020
- Faculty Supervisor of Undergraduate Senior Thesis in Generative Adversarial Networks for Noise Reduction within RF-based Biomedical Sensors, 2019
- Faculty Supervisor for 2 CS undergraduate students collaborating on an application project in Real-Time Visualization of RF-based Localization Systems, 2019
- Faculty Mentor to 4 STAR undergraduate research students, and Co-Faculty Mentor to 5 additional STAR students, 2 VIP students, 2 RET high school teachers, and 1 Honors Option from CS370 Operating Systems, for Biomedical Textile Analytics, Sensor Fusion, and Energy Harvested Device Actuation through RF, 2019
- Faculty Supervisor of an undergraduate research Co-Op in High Performance Computing to Parallelize Biomedical Imaging Machine Learning Processes for Digital Pathology, Summer 2019
- Faculty Mentor to 6 VIP students, including STAR Scholars, for Biomedical Textile Analytics experiential and research education projects, 2018
- Faculty supervisor of two undergraduate co-op students in Biomedical Analytics under CURE grant, 2017-2018; one of whom was named a 2018 Goldwater Scholar
- Co-Faculty Mentor of Research Experiences for Teachers (RET) project in Biomedical Multisensor Data Fusion, 2016-2017 and 2017-2018, and 2020 Summer Workshop
- Technical Editor for Ethical Case Studies for NSF EESE: The Ethics of Algorithms, 2016-2017
- Faculty supervisor for two students under an NSF-Sponsored Research Experiences for Undergraduates (REU) Supplement, 2016
- Co-Faculty Mentor of Students Tackling Advanced Research (STAR) freshman research student in Biomedical Multisensor Data Fusion
- Co-Advisor for CS and ECE senior design group project: Biomedical smart actuation
- Co-Advisor for CS, Mechanical Engineering, and ECE senior design group project: Actuation for Deep-Venous Thrombosis
- Co-Advisor for ECE senior design group project: Software Infrastructure for Secure and Scalable Medical Sensor Networks, 2015-16
- Co-Faculty Mentor of Students Tackling Advanced Research (STAR) freshman research student in Biomedical Smart Textile Analytics
- Advisor for ECE senior design group project: Smart Fabric Monitoring Devices
- Faculty co-Sponsor of Masters in Medical Science (MMS) Student Projects: *Maternity*

“SMART Fabric Bellyband” to Monitor Uterine Activity and Assess Fetal Well-Being and Smart Onesie: Early Warning System for the Prevention of Sudden Infant Death Syndrome (SIDS)

- Co-Faculty Mentor of Students Tackling Advanced Research (STAR) freshman research students (2) in Distributed MapReduce with Hadoop
- Co-Founder of the Project Nebula Big Data Research Group
- Co-Faculty Mentor of Students Tackling Advanced Research (STAR) freshman research students (2) in Big Data and Analytics
- Faculty Mentor of Students Tackling Advanced Research (STAR) freshman research student group in Biomedical Systems
- Co-Advisor for CS senior design group project: REportal service integration
- Support external course projects in Circuit Design and Operating System Internals

ACADEMIC SUPERVISION

- **Ursinus College**
 - Faculty Coordinator for MATH/CS/STAT 350 Oral Presentations, Spring, 2026
 - Member of PhD Dissertation Exam Committee for Drexel University College of Engineering, 2026
 - Member of PhD Proposal Exam Committee for Drexel University College of Engineering, 2026
 - Member of PhD Dissertation Exam Committee for Drexel University College of Engineering, 2024
 - Member of PhD Proposal Exam Committee for Drexel University College of Engineering, 2024
 - Faculty Coordinator for MATH/CS/STAT 350 Oral Presentations, Spring, 2024
 - Member of PhD Dissertation Exam Committee for Drexel University College of Engineering, 2023
 - Faculty Coordinator for MATH/CS 350 Oral Presentations, Fall, 2022
 - Member of PhD Candidacy Exam Committee for Drexel University College of Engineering, 2023
 - Member of PhD Candidacy Exam Committee for Drexel University College of Engineering, 2023
 - Member of PhD Proposal Exam Committee for Drexel University College of Engineering, 2022
- **Drexel University**
 - Member of PhD Candidacy Exam Committee for College of Engineering, 2019-2020
 - Member of PhD Thesis Advisory Committee for College of Arts and Sciences, 2018-2019
 - Member of PhD Candidacy Exam Committee for College of Engineering, 2018-2019
 - Faculty stakeholder to a CS senior design group project EtherFx for software integration, 2018-2019
 - External Stakeholder for Smart Home Senior Design Project
 - Technical co-Advisor for Which MBA 2016, sponsored by the Economist: “Kaspersky Lab Cyber Security Case Study”
 - Advisor for CS senior design group project: Scan Technologies
 - External Stakeholder for Flight Control and Failure Detection Senior Design Project
 - Faculty Advisor for DUCSTeach education outreach student organization, 2016-2017
 - Advisor for Life Insurance Big Data Senior Design Project
 - Co-Advisor of Freshman Design Project: Big Data and Analytics using the Mathforum at Drexel University
 - Faculty Advisor to the Math and Computer Science (MCS) Society, ACM undergraduate student chapter, 2012-2015
 - Faculty Advisor to the Arduino Developers at Drexel University
 - Founder of the Web Services User Group at Drexel University
 - Advisor for Senior Design Group: Dynamic Traffic Control System
 - Advisor for Freshman Design Group: Drexel Map Android Application
 - Co-Advisor for CS senior design group project: Classroom Application Media Center

(CRAM)

- Advisor for ECE senior design group project: Alzheimer's In-Home System
- Advisor for CS senior design group project: Project Sekai - Experiencing the World
- Faculty Advisor to an IEEE Systems Student Competition team
- Advisor for ECE senior design group project: AutoCaddy (Autonomous Golf Cart)
- Technical Advisor for CS senior design group project: ANFS (Advanced NFS)
- Advisor for engineering freshman design group project: Interactive Android Battlefield Map
- Co-Advisor for CS senior design group project: Visual VHDL (V2HDL)

EDUCATIONAL INITIATIVES

- **Ursinus College**
 - Faculty lead of AI for All workshop for professional staff (two presentations), 2026
 - Faculty lead of AI for All workshops for faculty, 2025
 - Faculty lead of Problem-Based Learning (PBL) workshop series, 2025
 - Faculty lead of PBL Learning Circle, 2025
 - Instructor for WR Crigler Program and Summer Experience for Incoming First-Year Students on "Computing in Context," August, 2022
 - Guest Lecturer with Kelly Joyce in SOC313 (Sociology of Global Health) at Drexel University on Integrating Machine Learning and Sociology of Health under an award from the Colonial Academic Alliance, 2022
 - Co-Facilitator (with Dr. Jeffrey L. Popyack) of Pixel Pandemonium student activity for National Computer Science Education Week in Philadelphia, 2021, 2022
- **Drexel University**
 - Instructor at CodeFest for HS Students at Philly CodeFest, May, 2019
 - Facilitator for Teaching Assistant Training in Inclusive Pedagogy, 2019
 - Piloted peer mentorship program in CI101 for approximately 80 freshman students with support from the Howard Hughes Medical Institute and the Drexel University College of Arts and Sciences, 2017-2018
 - Instructor for the Drexel University Computing Academy (DUCA) High School Summer Program and (formerly) the Pennsylvania Governor's School in Information, Society and Technology (PGSIST), 2008-2015
 - Student Teacher in Grade 8 Mathematics at Middle Years Alternative (MYA), 2007-2008
 - Guest Lecturer in AP Computer Science at Upper Darby High School, 2002-2004

FACULTY SERVICE INSTITUTIONAL SERVICE INITIATIVES

- **Ursinus College**
 - Member of the AI Advisory Task Force, 2025-Present
 - Faculty Advisor of the Robotics Club, 2025-Present
 - Member of the Retention Advisory Group, 2025-Present
 - Member of Faculty Council, 2025-2029
 - Interim Member of Faculty Council (*ex officio* Interim Faculty Meetings Chair), 2025-2026
 - Member of Faculty Council (*ex officio* Faculty Meetings Chair), 2026-2029
 - Member of the Faculty Task Force on Academic Program Evaluation for the Sciences Division, 2025
 - Interim Member of the Institutional Assessment Committee, 2026
 - Faculty Meetings Chair, 2026-2029
 - Interim Faculty Meetings Chair, 2026
 - Elected Faculty Meetings Chair, 2026-2029
 - Member of the Core Outcomes Assessment Committee, 2025-Present
 - Developed software system for AI-enabled core student outcomes assessment, 2025-2026
 - Faculty Advisor of the Badminton Club, 2025-Present
 - Faculty Representative on the College Board of Trustees Committee on Student Experience, 2025-2026
 - Member of the College Intellectual Property Committee, 2025-Present

- Pre-Engineering Program Advisor for Computer Science, 2025-Present
- Member of the APEX Implementation Working Group, 2025-Present
- Member of the Art & Science Initiative Working Group on Curriculum, co-Curriculum, Wellness, and Student Life, 2024-2025
- Faculty Representative on the College Board of Trustees Committee on Advancement, 2024-2025
- Faculty Judge for ENTR-120 “Startups and the Scientific Method” student projects, 2024, 2025
- Faculty Participant in the *Entrepreneurial Humanities* Learning Circle, 2024-2025
- Faculty Advisor for the Ursinus Chapter of Upsilon Pi Epsilon: International Honor Society for the Computing and Information Disciplines, 2024-Present
- Member of the First Year Advising Committee, 2023-2025
- Member of the Vice President for Finance and Administration (VPFA) Search Committee, 2023-2024
- Parliamentarian, 2023-2026
- Faculty Search Interviewer for Business and Economics Tenure Track Search, 2022-2023
- Faculty Participant in External Department Review for Physics Department, 2022-2023
- First-Year Academic Advisor, 2022-Present
- Major Academic Advisor, 2022-Present
- Member of the Midterm Academic Intervention Committee, 2022-Present
- Early Alert Focus Group Member, 2021-2023
- **Drexel University**
 - Panelist for First Year Exploratory Career Panel, 2019-2020
 - Chair of the CCI Faculty Promotion Committee, 2018-2019
 - Member of the CS Department Head Search Committee, 2018-2019
 - Member of the CCI Corporate Partners Program Committee, 2018-2019
 - Member of the Provost’s DC Programming Summit, 2018
 - Cohort participant in Drexel Leaders 2020 professional development program, 2017
 - Member of the Faculty Senate Budget Planning and Development (BP&D) Committee, 2017-2018
 - Member of the Isaac L. Auerbach Innovation Award Committee, 2017-2018
 - Member of the Faculty Data and Assessment Committee, Chair of the Survey Baseline Data Group Subcommittee, 2017-2019
 - Faculty liaison for the Steinbright Career Development Center (SCDC) Employment Summary and Planner (ES&P) experiential curriculum integration focus groups for returning co-op students, 2017-2019
 - Member of the Steinbright Faculty Advisory Board, 2017-2019
 - Member of the University Program Alignment and Review (PAR) Committee, 2016-2019
 - Member of the CCI Women in Computing (WiC) initiative, 2016-2019
 - Member of the Center for the Advancement of STEM Teaching and Learning Excellence (CASTLE) network at Drexel University, 2016-2019; Honest Data Broker for experiential learning research, 2018-2019
 - Member of University Advising Implementation Committee (UAIC) on Student Major Changes across the University, 2016-2018
 - Member of the CCI Undergraduate Data Science Curriculum Committee, 2016-2018
 - Member of the Dean Search Committee for the College of Nursing and Health Professions (CNHP), representing the Faculty Senate, 2016-2017
 - Member of the Philly Codefest Advisory Board, 2016-2017
 - Member of the University Advising Steering Committee and Chair of Advising Infrastructure and Workflow Subcommittee, 2016-2017
 - Member of the CCI Faculty/Academic Council, 2015-2016
 - Member of the CCI College Curriculum Committee 2015-2019, co-Chair 2016-2017, 2018-2019
 - Member of the Academic Standing Committee, 2015-2019
 - Chair of the CCI Program Alignment and Review (PAR) Committee, 2015-2016, 2019-2020
 - Member (interim) of the CCI Finance and Strategic Planning Committee, 2015
 - Member (interim) of the CCI Bylaws Committee, 2015

- Member (interim) of the CCI Executive Council, 2015
- Acting Department Head August, 2015-January, 2016
- University Faculty Senator from CCI, 2014-2017
- Recording Secretary for the CCI Senate Caucus, 2014-2017
- Member of the CCI Computing Faculty Resources Committee, 2014-2016
- Member of the First-Year CCI Curriculum Committee, 2014-2015
- Member of the CCI Faculty Search Committee, 2013-2015
- Chair of CCI Computing Assessment and ABET Accreditation Committee, and Program Assessment Coordinator for all undergraduate programs and concentrations within Computer Science and Software Engineering, 2012-2016
- Member of the College of Engineering (COE) Assessment Executive Committee, 2012-2014
- Chair of the CS Department Assessment Committee, 2011-2019
- Member of the COE Assessment Committee, 2011-2012
- Member of the CS and SE Undergraduate Computing Curriculum Committee, 2009-2019, Chair 2015-2019
- Developed a new track and curriculum for the CS Architecture Track concentration, 2008-2011
- Chair of the CS Undergraduate Retention / Core Curriculum Committee, 2008-2014

COMMUNITY SERVICE INITIATIVES

- **Ursinus College**
 - Founder and Trustee of the Ursinus Collegeville Amateur Radio Club (N3URS), 2025
 - Reviewer for CSTA K-12 Standards for Computer Science, 2025-2026
 - Review Committee Member for CS4Philly Computer Science Teacher Awards, 2025-2026
 - Co-coordinator for Remake Learning Day HACK Club Hackathon at Avon Grove High School, 2024
 - External Reviewer for Computer Science at Wabash College, 2024
 - Faculty participant in NCWIT Learning Circles, 2023-2024
 - Faculty participant in CRA LEVEL UP Workshop, 2023
 - Emcee for Vex Robotics and Drone Qualifying Competitions for Chester and Delaware County Teams at Avon Grove High School, 2023-2026
 - Coordinator and judge for CSTA Longwood Hackathon at West Chester University, 2023
 - Judge for “Shark Tank” AP Computer Science Principles design projects at Avon Grove High School, 2023-2025
 - Guest Lecturer for Programming 1 and 2 classes at Avon Grove High School, 2023-2025
 - Member of the Philadelphia STEM Ecosystem, 2023-Present
 - Member of the Citizen Science for Community Service Planning Committee, 2026-Present
 - External Stakeholder Group Member for STEAM Curriculum Development at the School District of Philadelphia, 2022
 - Faculty facilitator for CSTA Longwood Drone Event at Avon Grove High School, September, 2022
 - Focus Group Participant for the Philadelphia STEM Equity Collective on Expanding Access to STEM Career Pathways, May, 2022
 - Faculty Judge for Research Experiences for Undergraduates (REU) at Temple University, 2021, 2022
 - Judge for PA SEED Hackathon on Environmental Literacy at the Delaware County Intermediate Unit (DCIU), January, 2020
- **Drexel University**
 - Moderator for Center for the Integration of Research, Teaching and Learning (CIRTL) Conference in Philadelphia, PA, 2019
 - Faculty Judge for Research Experiences for Undergraduates (REU) at Temple University, 2019
 - Judge for Exelon STEM Academy for High School Girls, July, 2019
 - Judge for PA Governor’s STEM Design Competition for K-12, Delaware Valley Region, 2019, 2020

- Project Mentor and Judge for PA Department of Agriculture Seed K-12 projects on the Spotted Lanternfly with the Delaware County Intermediate Unit (DCIU), 2018
- Member of the Delaware County Intermediate Unit (DCIU) STEM Education Council, 2017-2019
- Mentor and Judge for Philly CodeFest Hackathon, 2015
- Judge for Students Tackling Advanced Research (STAR) Undergraduate Research Program
- Judge for Drexel University Research Day Competition

EDITORIAL SERVICE

- **Ursinus College**
 - Guest Editor for MDPI Special Issue on Machine Learning for Wireless Biomedical Applications, 2023

REFeree SERVICE

- **Ursinus College**
 - NSF Review Panelist for DUE, 2025
 - Reviewer for ACM SIGCSE, 2024
 - Reviewer for IEEE Sensors Journal, 2023
 - Reviewer for IEEE Journal of Biomedical and Health Informatics (JBHI), 2023
 - Reviewer for IEEE Journal of Biomedical and Health Informatics (JBHI), 2022
 - Reviewer for IEEE Signal Processing in Medicine and Biology (SPMB), 2020
 - Reviewer for IEEE Access Journal (three engagements), April, 2020 through May, 2020
- **Drexel University**
 - Reviewer for IEEE Sensors Journal, 2018
 - Reviewer for PLOS One Journal, 2018
 - Reviewer for Elsevier Pervasive and Mobile Computing, 2015
 - Reviewer for ACM SIGCSE, 2013
 - Reviewer for ACM SIGCSE, 2012
 - Reviewer for the International Advisory Committee for the International Joint Journal Conference in Engineering (IJJCE) 2009

CONFERENCE SERVICE

- **Ursinus College**
 - Organizing Committee Member and Session Leader for CSTA UnConference in Pennsylvania, 2023-Present
 - Technical Program Chair for IEEE SPMB, 2022
 - Program Chair for IEEE SPMB, 2021
 - Session Chair for IEEE SPMB, 2021
- **Drexel University**
 - Session Chair for ACM SIGCSE, 2012

TEXTBOOK AND CURRICULUM REVIEW SERVICE

- **Drexel University**
 - External Reviewer for CS401 Operating Systems Course, 2012
 - External Reviewer for CS402 Local Area Networks Course, 2012
 - External Reviewer for CS407 Network Apps Programming Course, 2012
 - External Reviewer for CS412 Mobile App Development, 2012
 - Reviewer for *Big Java Late Objects, 5th Edition*, 2012
 - Reviewer for *Computer Systems Architecture: Theme and Variations*, 2012
 - Reviewer for *Computer Systems: A Programmer's Perspective*, 2011
 - Reviewer for *Introduction to Internetworking, Its Security and Applications*, 2011

RESEARCH FUNDING AND SCHOLARLY DISSEMINATION *Note: Undergraduate students are denoted by *, K-12 teachers are denoted by †, and the corresponding/presenting author(s) are underlined, where appropriate. Selected manuscripts are available at <https://www.billmongan.com/publications>*

PATENTS

- P1 A Wearable and Portable Smart Actuation Device for DVT Risk Mitigation: Deep Vein Thrombosis Prevention Device (DVT-PD). Hassan El Mghari*, Gregory Olsen*, Zikang Ling*, Srivasta Ganesh*, William Mongan, Vasil Pano, and Kapil Dandekar. Provisional filed May 14, 2021 as US Patent Application No. 63/188,678 and published as US Patent WO2022241306A1 November 17, 2022; Patent Application No. 18/560,487 filed November 13, 2023 and published as US Patent US20240252389A1 August 1, 2024.

SOFTWARE PACKAGES AND DATASETS

W16 **mcp**proxy

- <https://github.com/BillJr99/mcp>proxy
- Config-driven MCP host exposing a single HTTP endpoint; tool providers defined as YAML files embedding Python `async` handler functions or delegating to npx MCP packages; built-in FastAPI web UI for provider editing, secret management, and live command streaming; Docker and pre-built GHCR image support.
- Flask, FastAPI; Python; 2026.

W15 **AutoGUI Desktop Agent**

- <https://github.com/BillJr99/AutoGUI>
- Vendor-neutral desktop automation agent connecting any OpenWebUI or Ollama-compatible LLM to OS-level controls via a ReAct-style agentic loop; supports shell, filesystem, accessibility-tree clicking (UIAutomation/AT-SPI), Set-of-Mark visual grounding, Playwright browser automation, skill library, and a FastAPI REST server.
- Experimental prototype; Python, TypeScript; 2026.

W14 **BetterWebUI**

- <https://github.com/BillJr99/BetterWebUI>
- Faculty-friendly agentic front end for OpenWebUI; adds workspaces, on-demand skills, MCP server management, CLI shortcuts, and KaTeX math rendering, with an approval-gated shell and integrations for CognitiveLoopKernel, AutoGUI, and OSScreenObserver.
- Python, FastAPI, HTML/CSS/JavaScript; 2026.

W13 **Cognitive Loop Kernel (CLK)**

- <https://github.com/BillJr99/CognitiveLoopKernel>
- Local-first multi-agent development harness that iterates a natural-language idea into a working software system through dynamic agent casting, YAML workflow orchestration, and automatic git commits.
- Provider-agnostic (Claude Code, Codex, Gemini, Ollama, pi, OpenWebUI); Python, TypeScript, Shell; 2026.

W12 **OS Screen Observer**

- <https://github.com/BillJr99/OSScreenObserver>
- Cross-platform Python prototype exposing the OS accessibility tree, OCR, and vision-model descriptions through a browser dashboard and an MCP stdio server for AI agent integration.
- Supports Windows (UIA), macOS (AXUIElement), and Linux (AT-SPI); Python; 2026.

W11 **pi-openai-compat**

- <https://github.com/BillJr99/pi-openai-compat>
- TypeScript extension for the pi terminal harness registering OpenAI-compatible providers (OpenRouter, NVIDIA NIM, Ollama, custom) as native models in pi's `/model` picker.
- Published to npm (@billjr99/pi-openai-compat); MIT License; 2026.

W10 **llm**proxy

- <https://github.com/BillJr99/llm>proxy
- OpenAI-compatible HTTP proxy aggregating multiple LLM providers behind a single endpoint; model routing via provider-prefix naming; hot-reloaded configuration; Docker support.
- Flask, gunicorn; Python; 2026.

W9 **digital link**

- https://github.com/BillJr99/digital_link
- Shell script enabling DTMF-controlled switching between digital voice modes (DMR, D-STAR, YSF, NXDN, P25) and talkgroups on AllStarLink nodes running DVSwitch via Asterisk.
- Shell; 2025.

W8 pxt-ESP8266_VarIOT

- https://github.com/BillJr99/pxt-ESP8266_VarIOT
- MakeCode extension for the BBC micro:bit enabling WiFi telemetry upload to a VarIOT ThingsBoard IoT gateway via an ESP8266 module; used in K-12 and undergraduate IoT instruction.
- TypeScript, MakeCode; 2024.

W7 Ursinus WebIDE

- <https://billmongan.com/Ursinus-WebIDE>
- Browser-based integrated development environment for introductory and intermediate computing coursework, supporting in-browser code execution, automated test case evaluation, per-exercise instructor configuration, and multi-section course management; co-developed with Christopher Tralie (Ursinus College).
- Entirely client-side static deployment: student code executes locally in the browser with no server infrastructure required, enabling zero-cost public access and equity-oriented support for Chromebooks and students with constrained computing environments.
- Provides automated mistake detection with targeted, hint-bearing feedback; a virtual filesystem experience that scaffolds local development conventions; RSA-encrypted, FERPA-compliant grade submission to Canvas; and multimedia execution support including audio synthesis, matplotlib visualization, and WebGL shader programming.
- Supports Java, Python, JavaScript, C++, SQL, Scheme, Prolog, R, and Pyodide; JavaScript, HTML/CSS; 2020.

W6 IoT Data Collection Framework

- <https://github.com/drexelwireless/iot-sensor-framework>
- Modular Python framework for real-time collection and storage of heterogeneous IoT sensor data, including UHF RFID interrogators, with configurable driver and backend support.
- Associated with Ph.D. Dissertation, Drexel University, 2018; Python.
- Public release v1.0, May 2020; DOI: <https://doi.org/10.5281/zenodo.3786932>.

W5 IoT Data Processing Framework

- <https://github.com/drexelwireless/iot-processing-framework>
- Python suite for real-time visualization and ML experimentation on IoT sensor data with sensor fusion, ground truth comparison, and data perturbation modules.
- Associated with Ph.D. Dissertation, Drexel University, 2018; Python.
- Public release v1.1, May 2020; DOI: <https://doi.org/10.5281/zenodo.3786930>.

W4 Bellyband Datasets

- <https://github.com/drexelwireless/bellyband-datasets>
- Public dataset of RFID-based respiratory and biomedical measurements from simulated mannequins and emulated environments using the passive smart garment bellyband device.
- Companion dataset to the IoT Sensor and Processing Frameworks.
- Associated with Ph.D. Dissertation, Drexel University, 2018; Python.

W3 REportal 2.0

- <https://www.billmongan.com/software/reportal/>
- Service-Oriented Architecture reverse engineering portal integrating static and dynamic analysis tools (Bunch, Chava, Ciao) through a unified web interface with automatic Aspect instrumentation.
- Associated with MS Thesis, Drexel University, 2008; Java/SOA.

W2 ClusterNav

- <https://www.billmongan.com/software/clusternav/>
- Java GXL visualizer for software module dependency graphs produced by Bunch, supporting interactive expansion and collapse of clusters at varying levels of abstraction.
- Integrated into REportal 2.0; Java.

W1 **xml2dot**

- <https://www.billmongan.com/software/xml2dot/>
- Java tool for visualizing the structure of XML documents using GraphViz; developed originally for inspection of Impinj RFID data reports.
- Java.

GRANT FUNDING

- G22 Expanding Computing Education Pathways (ECEP) Co-Sponsorship award. co-Principal Investigator with Sara Frey. ECEP Alliance and NSF, 2025, \$15,000.
- G21 MENTIR-AI: AI Analysis and Feedback of Students' Mathematical Thinking. co-Principal Investigator with Jason Silverman, Principal Investigator, Annie Fetter, Valerie Klein, Amanda Reinsburrow, Wesley Shumar, Stephen Weimar, Cindy Yovanov, co-Principal Investigators. Gates Foundation, 2025-2026, \$499,806.
- G20 NSF S-STEM (DUE-2527624): PERSIST: Partnerships for Engagement, Resilience, and Support through Interdisciplinary STEM Teamwork. Principal Investigator with Mark Ellison, Casey Schwarz, Lisa Grossbauer, and Carlita Favero, co-Principal Investigators. 2025-2031, \$1,999,659.
- G19 Online Coding Modules for Low-Stakes Student Exercises in Computing. Co-Principal Investigator with Chris Tralie (Principal Investigator). Cutler Grant for Open Educational Resources (OER) Development, 2024-2025, \$1,000.
- G18 Special Topics First Year Advising Sections. Co-Principal Investigator with Katie O'Brien (Principal Investigator) and Lew Riley. 2024-2025, \$2,500.
- G17 Demystifying Hidden Curriculum: Realizing the Dream of Egalitarian Teaching and Learning. Project Lead with Meredith Goldsmith and Lew Riley, *et al.* Ursinus College Innovation Council (CIC) Proposal, 2023-2024, \$1,000.
- G16 NIH Smart and Connected Health (SCH) (R01EB029364): Smart and Connected Health for Newborn Ventilation (co-Principal Investigator with Kapil R. Dandekar, Principal Investigator, and Vineet Bhandari, Anup Das, Genevieve Dion, co-Principal Investigators), 2019-2023, \$1,038,686.
- G15 Internet of Things for Future Smart Campus and City (co-Principal Investigator with Kapil R. Dandekar, Principal Investigator, *et al.*), 2019-2020; Drexel DARE Proposal, \$250,000 (estimated).
- G14 NSF CyberCorps Scholarship for Service (SfS) Site (DGE-1922202) for Undergraduate Education in Cyber Security (co-Principal Investigator with Brian Smith, Ayana Allen-Handy, Thomas Heverin, Michelle Rogers, Toni Sondergeld, co-Principal Investigators), 2019-2024, \$3,999,970.
- G13 NSF Computer and Network Systems (CNS): NeTS: Small (award number CNS-1816387): Architectures and Testbed for Functional Fabric RFID in the Internet of Things (co-Principal Investigator with Kapil R. Dandekar, Steven Weber, Genevieve Dion, co-Principal Investigators), 2018-2022, \$499,647. REU Supplement to support two undergraduate students, \$16,000 estimated, 2021.
- G12 Arthur Vining Davis Foundations: Research on Experiential STEM Curricula for Authentic Learning Experiences (RESCALE) (co-Principal Investigator with Jennifer Stanford, Jason Silverman, Adam Fontecchio, Eric Brewe, co-Principal Investigators), 2018-2019, \$274,998.
- G11 Colonial Academic Alliance IN/CO: Tracking Experiential Learning Outcomes Across Three CAA Campuses (co-Principal Investigator with N. John DiNardo, Adam Fontecchio, Jennifer Stanford, *et al.*, co-Principal Investigators), 2017-2019, \$40,000.
- G10 NSF Networking Technology and Systems (NeTS), **E**Arly-concept **G**rants for **E**xploratory **R**esearch (EAGER) (award number CNS-1738070): SC2: Team Dragon Radio (co-Principal Investigator with Kapil Dandekar, Principal Investigator, Steven Weber, Nagarajan Kandasamy, and Geoffrey Mainland, co-Principal Investigators), 2017, \$99,978.

- G9 Analytics on Real-Time Biometrics from Passive Wearable Smart-Garments (co-Principal Investigator with Adam Fontecchio, Principal Investigator); 2017-2018, Commonwealth Universal Research Enhancement (CURE) Formula Grant (SAP117558-014), \$75,000. Drexel University Co-op Funding Award supplement of \$7,250 to support undergraduate experiential learning in research.
- G8 Integrated Design Research and Engineering for Advanced Manufacturing (iDREAM) (Senior Personnel with Genevieve Dion, *et al*), 2016; Drexel DARE Proposal, \$250,000.
- G7 Experiential Learning through the Cooperative Education Lifecycle (ExCEL) (Senior Personnel with Adam Fontecchio, Jason Silverman, Jennifer Stanford, Pramod Abichandani, Kapil Dandekar, David Goldberg, Antonios Kontsos, Suzanne Rocheleau, and Brian Smith), 2016; Drexel DARE Proposal, \$250,000.
- G6 Three Proosals for Student-Centered Spaces for Computing and Informatics. Bruce Char and William Mongan. Drexel University Steelcase Award, 2015.
- G5 NSF/IEEE-TCPP Curriculum Initiative on Parallel and Distributed Computing – Core Topics for Undergraduates: Using BigData for Learning about a Slice of Parallel Computation in Several Courses (Co-Principal Investigator with Bruce Char and Jeffrey Popyack), July, 2015-July, 2017. \$2,500 faculty development award.
- G4 NIH CPS: TTP Option: Synergy: Sensing, Processing, and Actuation of Biomedical Smart Textiles for Deep Venous Thrombosis Prevention (award number U01EB023035), March, 2016-March, 2020 (co-Principal Investigator with Kapil Dandekar, *et al*), \$1,392,557 (estimated).
- G3 NSF Partnerships For Innovation: Building Innovation Capacity (PFI:BIC) Grant Number IIP-1430212 on Biomedical Smart Textiles, 2014-2017 (Senior Personnel with Kapil Dandekar, *et al*, co-Principal Investigators), \$799,577. REU Supplement to support two undergraduate students, \$16,000 estimated, 2016.
- G2 IBM Big Data and Analytics Education Grant, 2013 (Jeffrey L. Popyack and William M. Mongan, co-Principal Investigators), \$10,000.
- G1 NSF Research Experiences for Teachers (RET) in Engineering and Computer Science Site for Machine Learning, Big Data and CS Principles, National Science Foundation, DUE-0837665, July 2013-June 2016 (Senior Personnel and co-Director with Jeffrey L. Popyack, Principal Investigator, Mary Jo Grdina, co-Principal Investigator), \$499,990.

DISSERTATION AND THESIS

- D2 William M. Mongan. **Predictive Analytics on Real-Time Biofeedback for Actionable Classification of Activity State**. PhD Dissertation, Drexel University, 2018.
- D1 William M. Mongan. **A Service-Based Web Portal for Integrated Reverse Engineering and Program Comprehension**. MS Thesis, Drexel University, 2008.

BOOK CHAPTERS

- B2 Chelsea Amanatides, Stephen Hansen*, Ariana S. Levitt, Yuqiao Liu, Patrick O'Neill*, Damiano Patron, Robert Ross*, Daniel Schwartz, Jesse Stover*, Md Abu Saleh Tajin, Genevieve Dion, Adam K. Fontecchio, Vasil Pano, William M. Mongan, and Kapil R. Dandekar. **Wearable Smart Garment Devices for Passive Biomedical Monitoring**. *Biomedical Signal Processing*, Volume 2, Chapter 4. Iyad Obeid, Ivan Selesnick, and Joseph Picone, eds, Springer: April, 2021.
- B1 Duc N. Nguyen, Kyle Usbeck, William M. Mongan, Christopher T. Cannon, Robert N. Lass, Jeff Salvage, William C. Regli, Israel Mayk, Todd Urness. **A Methodology for Developing an Agent Systems Reference Architecture**. *Agent-Oriented Software Engineering XI*, pp. 177-188. Danny Weyns, Marie-Pierre Gleizes, eds, Springer Berlin Heidelberg: 2011.

ARCHIVAL JOURNAL PUBLICATIONS

- J8 Md Abu Saleh Tajin, Md Shakir Hossain, William M. Mongan, and Kapil R. Dandekar. **Passive UHF RFID-based Real-Time Intravenous Fluid Level Sensor.** *IEEE Sensors Journal*, 2023.
- J7 Ankita Paul, Md Abu Saleh Tajin, Anup Das, William M. Mongan, and Kapil R. Dandekar. **Energy-Efficient Respiratory Anomaly Detection in Premature Newborn Infants.** *MDPI Electronics Journal Special Issue: Neuromorphic Sensing and Computing Systems*, 2022.
- J6 Md Abu Saleh Tajin, Marko Jacovic, Genevieve Dion, William M. Mongan, and Kapil R. Dandekar. **UHF RFID Channel Emulation Testbed for Wireless IoT Systems.** *IEEE Access Journal*, 2021.
- J5 Md Abu Saleh Tajin, William M. Mongan, and Kapil R. Dandekar. **Passive RFID-based Diaper Moisture Sensor.** *IEEE Sensors Journal*, October 2020.
- J4 Sayandeep Acharya, William M. Mongan, Ilhaan Rasheed, Yuqiao Liu, Endla Anday, Genevieve Dion, Adam Fontecchio, Timothy Kurzweg, and Kapil R. Dandekar. **Ensemble Learning Approach via Kalman Filtering for a Passive Wearable Respiratory Monitor.** *IEEE Transactions of Biomedical and Health Informatics*, May 2019.
- J3 Damiano Patron, William Mongan, Timothy Kurzweg, Adam Fontecchio, Genevieve Dion, Endla Anday, and Kapil R. Dandekar. **On the Use of Knitted Antennas and Inductively Coupled RFID Tags for Wearable Applications.** *IEEE Transactions on Biomedical Circuits and Systems*, January 2016.
- J2 William Regli, Israel Mayk, Christopher Cannon, Joseph Kopena, Robert Lass and William M. Mongan. **Development and Specification of a Reference Architecture for Agent-Based Systems.** Published in the *IEEE Transactions on Human-Machine Systems*, 2013.
- J1 William Regli, Israel Mayk, Christopher J. Dugan, Joseph B. Kopena, Robert N. Lass, Pragnesh Jay Modi, William M. Mongan, Jeff K. Salvage and Evan A. Sultanik. **Development and Specification of a Reference Model for Agent-Based Systems.** Published in *IEEE Transactions on Systems, Man, and Cybernetics*, September 2009.

CONFERENCE AND WORKSHOP PROCEEDINGS PUBLICATIONS

- C19 Amanda Reinsburrow, Cindy Yovanov, William M. Mongan, Michael Cummins*, Matthew Latta*, Wesley Shumar, Valerie Klein, Jason Silverman, and Steve Weimar. **Instructional Judgment, Not Just Correctness: Evaluating AI-Generated Feedback on Student Mathematical Thinking.** *Society for Information Technology and Teacher Education (SITE)*, March, 2026.
- C18 Robert Ross*, William M. Mongan, Patrick O'Neill*, Ilhaan Rasheed, Adam Fontecchio, Genevieve Dion, and Kapil R. Dandekar. **An Adaptively Parameterized Algorithm Estimating Respiratory Rate from a Passive Wearable RFID Smart Garment.** *IEEE Symposium on Mobile, Wearable and Ubiquitous Computing at the IEEE Conference on Computers, Software, and Applications (COMPSAC)*, July, 2021 **Acceptance Rate 27%**
- C17 Md Abu Saleh Tajin, Marko Jacovic, Xaime Rivas Rey, William M. Mongan, and Kapil R. Dandekar. **Channel Emulation for the Characterization of Wearable RFID Antennas.** *21st Annual IEEE Wireless and Microwave Technology Conference (WAMICON 2020-2021)*, April, 2021.
- C16 Stephen Hansen*, Daniel Schwartz, Jesse Stover*, Md Abu Saleh Tajin, William M. Mongan, and Kapil R. Dandekar. **Fusion Learning on Multiple-Tag RFID Measurements for Respiratory Rate Monitoring.** *IEEE International Conference on Bioinformatics and Biomedical Engineering (BIBE)*, October, 2020.
- C15 Patrick O'Neill*, William M. Mongan, Robert Ross*, Sayandeep Acharya, Adam K. Fontecchio, and Kapil R. Dandekar. **An Adaptive Search Algorithm for Detecting Respiratory Artifacts Using a Wireless Passive Wearable Device.** *IEEE Signal Processing in Medicine and Biology (SPMB)*, December, 2019.

- C14 Austin Gentry, William M. Mongan, Brent Lee*, Owen Montgomery, and Kapil Dandekar. **Activity Segmentation Using Wearable Sensors for DVT/PE Risk Detection.** *The First IEEE International Workshop on Integrated Smart Healthcare (WISH 2019) at the IEEE Conference on Computers, Software, and Applications (COMPSAC), July, 2019.*
- C13 William M. Mongan, Robert Ross*, Ilhaan Rasheed, Yuqiao Liu, Khyati Ved, Endla Anday, Kapil Dandekar, Genevieve Dion, Timothy Kurzweg, and Adam Fontecchio. **Data Fusion of Single-Tag RFID Measurements for Respiratory Rate Monitoring.** *IEEE Signal Processing in Medicine and Biology (SPMB) 2017, December, 2017.*
- C12 Shrenik A. Vora, William M. Mongan, Endla K. Anday, Kapil R. Dandekar, Genevieve Dion, Adam K. Fontecchio, and Timothy P. Kurzweg. **On Implementing an Unconventional Infant Vital Signs Monitor with Passive RFID Tags.** *IEEE International Conference on RFID, 2017, May, 2017.*
- C11 William Mongan, Ilhaan Rasheed, Khyati Ved, Shrenik Vora, Kapil Dandekar, Genevieve Dion, Timothy Kurzweg, and Adam Fontecchio. **On the Use of Radio Frequency Identification for Continuous Biomedical Monitoring.** *ACM/IEEE International Conference on Internet-of-Things Design and Implementation (IoTDI) 2017, April, 2017.*
- C10 William M. Mongan, Ilhaan Rasheed, Khyati Ved, Ariana Levitt, Endla Anday, Kapil Dandekar, Genevieve Dion, Timothy Kurzweg, and Adam Fontecchio. **Real-Time Detection of Apnea via Signal Processing of Time-Series Properties of RFID-Based Smart Garments.** *IEEE Signal Processing in Medicine and Biology (SPMB) 2016, December, 2016.*
- C9 William Mongan, Endla Anday, Genevieve Dion, Adam Fontecchio, Kelly Joyce, Tim Kurzweg, Yuqiao Liu, Owen Montgomery, Ilhaan Rasheed, Cem Sahin, Shrenik Vora, and Kapil Dandekar. **A Multi-Disciplinary Framework for Continuous Biomedical Monitoring Using Low-Power Passive RFID-based Wireless Wearable Sensors.** *Proceedings of the IEEE Smart Systems Workshop 2016, May, 2016.*
- C8 Shrenik Vora, William Mongan, Kapil Dandekar, Adam Fontecchio, and Tim Kurzweg. **Wireless Heart and Respiration Monitoring for Infants through Passive RFID Tags.** *International Conference on Biomedical and Health Informatics (BHI), February, 2016.*
- C7 Duc N. Nguyen, Robert N. Lass, Kyle Usbeck, William M. Mongan, Christopher T. Cannon, William C. Regli, Israel Mayk and Todd Urness. **Developing an Agent Systems Reference Architecture.** *Published in the Proceedings of the 11th International Workshop on Agent Oriented Software Engineering, May 2010.*
- C6 William M. Mongan and William C. Regli. **A Cyber-Infrastructure for Supporting K-12 Engineering Education through Robotics.** *Published in The Proceedings of the Association for the Advancement of Artificial Intelligence (AAAI) Education Track 2008.*
- C5 William M. Mongan, Maxim Shevertalov, Spiros Mancoridis. **Re-engineering a Reverse Engineering Portal to a Distributed SOA.** *Published in IEEE Proceedings of the 16th International Conference on Program Comprehension (ICPC) 2008.*
- C4 Quincy Brown, William Mongan, Elaine Garbarine, Dara Kusic, Eli Fromm, Adam Fontecchio. **Computer Aided Instruction as a Vehicle for Problem Solving: Scratch Programming Environment in the Middle Years Classroom.** *Published in Proceedings of the American Society for Engineering Education (ASEE) K-12 Track 2008.*
- C3 William M. Mongan, Christopher J. Dugan, Robert N. Lass, Andrew K. Hight*, Jeff Salvage, William C. Regli, Pragnesh J. Modi. **Dynamic Analysis of Agent Frameworks in Support of a Multiagent Systems Reference Model.** *Published in IADIS Proceedings of the International Conference Intelligent Systems and Agents (ISA) 2007.*
- C2 Christopher J. Dugan, Pragnesh Jay Modi, Joseph Kopena, William M. Mongan, William C. Regli, Israel Mayk. **A Reference Model for Agent-Based Command and Control Systems.** *Published in Proceedings of the 25th Army Science Conference 2006. Acceptance Rate 14%*
- C1 Pragnesh Jay Modi, Spiros Mancoridis, William M. Mongan, William Regli, Israel Mayk. **Towards a Reference Model for Intelligent Agent Systems.** *Published in Proceedings*

ABSTRACTS AND POSTERS

- A15 Christopher Tralie and William M. Mongan. **The Ursinus WebIDE: A Serverless Browser-Based Development Environment for Student Practice and Rapid Instructor Exercise Development.** *55th ACM Technical Symposium on Computer Science Education (SIGCSE TS 2026)*. St. Louis, MO, USA. February, 2026.
- A14 Ankita Paul, Md Abu Saleh Tajin, William M. Mongan, Anup Das, and Kapil R. Dandekar. **Energy Efficient Detection of Respiratory Anomaly using Spiking Neural Networks.** *IBM-IEEE AI Compute Symposium (AICS 2021)*. October, 2021. **Best Poster Award Winner**
- A13 **REThink CS @ Drexel: Bridging Connections in Philadelphia with a Research Experiences for Teachers Site.** *NSF EEC Grantees Conference with Jeffrey L. Popyack*. Washington, DC. October, 2019.
- A12 **Post-It Pandemonium: Teaching Image Representation and Compression with an “Unplugged” Activity.** *ACM SIGCSE 2019 Nifty Assignment with Jeffrey L. Popyack*. Minneapolis, MN. February, 2019.
- A11 **A Wearable Biomedical Monitoring System Enabled by the Internet-of-Things.** *NSF Connections in Smart Health (CSH) Workshop with Kapil R. Dandekar, Genevieve Dion, Adam Fontecchio, Timothy Kurzweg*. Charlottesville, VA. September, 2018.
- A10 **MapReduce Parallelism across the Curriculum: an Interim Report.** *8th NSF/TCPP Workshop on Parallel and Distributed Computing Education (EduPar-18) with Bruce Char and Jeffrey Popyack*. Vancouver, BC, Canada. May, 2018.
- A9 **Using BigData for Learning about a Slice of Parallel Computation in Several Courses.** *NSF/TCPP Workshop on Parallel and Distributed Computing Education (EduPar-16) Poster with Bruce Char and Jeffrey Popyack*. Chicago, IL. May, 2016.
- A8 **Pixels, Post-It’s and CS Principles.** *SIGCSE 2016 Poster with Jeffrey L. Popyack*. Memphis, TN. March, 2016.
- A7 **Statistical Analytics of Wearable Passive RFID-based Biomedical Textile Monitors for Real-Time State Classification.** *IEEE Signal Processing in Medicine and Biology (SPMB) Symposium Poster with Kapil Dandekar, Genevieve Dion, Tim Kurzweg, and Adam Fontecchio*. Philadelphia, PA. December, 2015.
- A6 **Raspberry HadoopPI: A Low-Cost, Hands-On Laboratory in Big Data and Analytics.** *SIGCSE 2015 Poster with Ken Fox* and Jeffrey L. Popyack*. Kansas City, MO, March, 2015 and *Drexel University Research Day 2015*.
- A5 **A Software Framework for Monitoring and Performing Analytics on Real Time Medical Device Data.** *American Society for Engineering Education Poster with Rachel M. Goeken*, Kapil Dandekar, Timothy Kurzweg, Genevieve Dion, and Adam K. Fontecchio*. Swarthmore, PA. November, 2014. **Runner-Up for Best Poster**
- A4 **Maternity Smart Fabric Bellyband to Monitor Uterine Activity and Assess Fetal Well-Being.** *Wearable Technology in Healthcare Society (WATCH) Conference with Kapil Dandekar, Genevieve Dion, Adam Fontecchio, Timothy Kurzweg, and Owen Montgomery, MD*. Indianapolis, IN. July, 2014.
- A3 **An Integrated Introduction to Network Protocols and Cryptography to High School Students.** *Poster Presentation at ACM SIGCSE 2012, Raleigh, NC*. March, 2012.
- A2 **GK-12: Engineering as a Contextual Vehicle for Science and Mathematics Education Poster.** *2007 NSF GK-12 Annual Meeting, Washington DC and 2007 Drexel Research (RISC) Day*.
- A1 **REportal: A Web-Based Reverse Engineering Portal.** *2003 Drexel University Research Day**.

STUDENT PRESENTATIONS

- S9 Levi Fritz*, Shannon Zura, and William M. Mongan. **Interactive AI Theater Presentation.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2026.
- S8 Michael Cummins*, Matt Latta*, Claire Shoemaker*, and William M. Mongan. **RAG vs. Fine-Tuning in Math Education: Developing and Validating Mentir AI.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2026.
- S7 Andrei Bogdan* and William M. Mongan. **Minute and Subject-Level Sleep Apnea Detection Using an ECG-Based Machine Learning Pipeline.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2026.
- S6 Michael Cummins* and William M. Mongan. **Developing Tools for AI Explainability.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2025.
- S5 Eugene Thompson*, George Psaradakis, and William M. Mongan. **Financial Tools for the Grizzly Guide to Wealth (GGTW).** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2025.
- S4 Arthur Artene*, Tristan Ashcroft, and William M. Mongan. **UCPlaces: Campus Orientation App Using ArcGIS.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2023.
- S3 Jonas Ling* and William M. Mongan. **RF IoT Security Layer.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2023.
- S2 Dylan Melby* and William M. Mongan. **Automated Moon Crater Detection: Exploring Algorithmic and Neural Network Approaches.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2023.
- S1 Kevin Hoffman* and William M. Mongan. **nViz: A Neural Network Visualizer.** *Ursinus Celebration of Student Achievement (CoSA)*. April, 2023.

TECHNICAL REPORTS

- R3 Israel Mayk and William C. Regli, eds; William M. Mongan, *et al.* **Agent Systems Reference Architecture.** Published 2008.
- R2 Israel Mayk and William C. Regli, eds; William M. Mongan, *et al.* **Agent Systems Reference Model.** Published 2006.
- R1 C. Liao, S. Mancoridis, W. Mongan*. **REportal: A Reverse Engineering Portal.** Drexel University Technical Report. Published 2003.

INVITED TALKS

- T40 **Rethinking the Schoolhouse with AI.** Invited Panelist. Collegeville, PA. December, 2025.
- T39 **AI: How Should We Live Together?.** Moderator and Panelist at *Ursinus College Family Weekend* with David Aipperspach, Scott King, and Kelly Sorensen. Collegeville, PA. September, 2025.
- T38 **AI in Advancement.** Keynote panelist at *Philadelphia Annual Fundraising Conference*, Collegeville, PA. July, 2025.
- T37 **Tools and Strategies for Leading Undergraduate Systemic Change.** *Learning Circles Panelist at NCWIT Summit 2024.* Kansas City, MO. May, 2024.
- T36 **Failure is not Failure: On the Importance of Exploration and Experimentation without Expectation in Teaching and Learning.** *Hidden Curriculum Series Presentation with Prof. Santiago Nunez-Corrales.* April, 2024.
- T35 **The State of AI in 2024: Understanding AI and How to Use it to Augment Your Life.** *Ursinus Alumni Tea Time.* March, 2024.
- T34 **The Technical Interview and Your Personal Brand.** *REU Site Workshop at Ursinus College.* June, 2023.
- T33 **Design Thinking as a Bridge from K-12 to the Workforce.** *PaTTAN CSinPA Virtual Community.* March-April, 2023.

- T32 **Equitable CS Education for Broader Workforce Preparation through Design Thinking.** *CS4All Summit Session at PaTTAN.* June, 2022.
- T31 **Introducing Computing Concepts through Multidisciplinary No-Code and Low-Code Platforms.** *CS4All Lunch and Learn Booth at PaTTAN with Jeffrey L. Popyack.* June, 2022.
- T30 **Broadening Computer Science Principles across the Curriculum and in Extracurricular Activities: Computing Pathways for K-12 Students.** *CS4All Summit Panel Moderator at PaTTAN with John Dougherty, Jamie Payton, Tammy Pirmann, and Jeffrey L. Popyack.* June, 2022.
- T29 **Professional Development to Highlight Pathways and Expand Access to the Computing Workforce from K-20.** *PaTTAN Remake Learning Days Monthly Meetup Moderator.* Harrisburg, PA. May, 2022.
- T28 **Replit in the Classroom.** *CSTA>>Philly Spring Symposium.* Philadelphia, PA. March, 2021.
- T27 **Teaching AP Computer Science A.** *Professional Development Workshop with the Delaware Department of Education.* Newark, DE. September, 2020.
- T26 **Grades 7-12 CS Bootcamp.** *2020 Pennsylvania Training and Technical Assistance Network (PATTAN) under a PASmart grant from the Pennsylvania Department of Education.* Harrisburg, PA. June, 2020.
- T25 **Using Github Classroom.** *CSTA>>Philly Spring Symposium.* Philadelphia, PA. April, 2020.
- T24 **Post-It Pandemonium: Teaching Image Representation and Compression with an “Unplugged” Activity.** *CS4Philly Workshop Activity with Jeffrey L. Popyack.* Philadelphia, PA. December, 2018.
- T23 **Contextualizing Principles of Computer Science.** *CASTLE Pedagogical Happy Hour.* Philadelphia, PA. May, 2018.
- T22 **CASTLE Summit at Drexel University.** *Faculty Panelist.* Philadelphia, PA. May, 2018.
- T21 **National Council of Women in Information Technology (NCWIT),** *Invited Faculty.* Tucson, AZ. May, 2017.
- T20 **Integrating Active Learning in the STEM Classroom (Inaugural Event).** *Pedagogical Readiness Oversight for Future Educators in STEM Subjects (PROFESS) at Drexel University.* Philadelphia, PA. April, 2017.
- T19 **A Panel Discussion on Computer Science Education.** *Computer Science Teachers Association Fall Symposium.* Villanova, PA. November, 2016.
- T18 **Wearable Technology Advances Health for Mothers and Babies.** *Yale Tech Summit presentation with Owen Montgomery.* New Haven, CT. October, 2016. Also presented to Teva in Frazer, PA, November, 2016.
- T17 **Wearable Smart Textiles Based on Programmable and Automated Knitting Technology for Biomedical and Sensor Actuation Applications.** *BIO International Convention with Kapil R. Dandekar, Genevieve Dion, Adam Fontecchio, Tim Kurzweg, Owen Montgomery, V.K. Narayan.* June, 2015.
- T16 **Feedback at Scale - Automatically Generated Feedback for CS Student Work: Best Practices.** *SIGCSE 2015 Birds-of-a-Feather Session with Bruce W. Char, Jeffrey L. Popyack, and Jeremy Johnson.* Kansas City, MO. March, 2015.
- T15 **Marconi: The Master of Wireless.** *Bellyband Demonstration at Drexel University.* Philadelphia, PA. October, 2014.
- T14 **Big Data, Big Deal; Welcome to the Twitterverse.** *Reboot, RETHink, Refresh with Jeff Popyack.* July, 2014.
- T13 **Big Data is Everywhere: Bridging Computing Disciplines and Society.** *Webinar for Drexel University with Jeff Popyack.* June, 2014 and January, 2015.

- T12 **What Is/Are CS Principles?** *Philadelphia Science Festival with Jeff Popyack and Omar Ali, School District of Philadelphia.* April, 2014.
- T11 **Learning Real-World Skills while Majoring in Computer Science.** *GETT: Girls Exploring Tomorrow's Technology with Jeff Popyack, Suzanne Hanbicki[†], and Hannah Pinkos*.* April, 2014.
- T10 **Internet Protocols and Cryptography.** *Reboot Renew RETHink Workshop at Drexel University.* July, 2013.
- T9 **Mobile Application Development with Web Services.** *Google CS4HS Workshop at the University of Pennsylvania.* August, 2012.
- T8 **Networking Applications, Protocols, and Cryptography.** *Computing Tapestry Workshop at the University of Pennsylvania.* July, 2012.
- T7 **Linux Kernel Vulnerabilities.** *Drexel University Math and Computer Science (MCS) Society Talk.* March, 2012.
- T6 **Networking Applications, Protocols, and Cryptography with Java.** *Google CS4HS Workshop at the University of Pennsylvania.* August, 2011.
- T5 **Version Control Systems.** *Drexel University Math and Computer Science (MCS) Society Talk.* July, 2010.
- T4 **Certification and standards for computing education in Pennsylvania.** Invited Panelist with Dougherty, J.P., Griffin, J., Pirmann, T., and Powell, R. Panel presentation submitted to *The Twenty-fifth Annual Consortium for Computing Science in Colleges (CCSC) Eastern Conference*, Villanova University, Villanova, PA. October 30 - 31, 2009.
- T3 **How the PC Starts its Day: From Boot Code to Boot Viruses.** *Drexel University Math and Computer Science (MCS) Society Talk.* February, 2009.
- T2 **Circuit Design with Breadboards, Microcontrollers, and FPGAs.** *Drexel University Math and Computer Science (MCS) Society Talk.* November, 2008.
- T1 **Computer Organization Workshop.** *The Pennsylvania Governor's Schools of Excellence: Information, Society & Technology.* Summer, 2008.

MEDIA COVERAGE

- M8 3 Main Line Area Colleges Offer Specialized Degrees in Artificial Intelligence.⁶ Lily Henderson. Main Line Today. April, 2026.
- M7 Ursinus Secures \$1.99M NSF Grant to Launch STEM Scholarship Program.⁷ Michael-Paul Kidd. Philadelphia Today. February, 2026.
- M6 Montgomery County Appoints Members to AI Advisory Council.⁸ Rachel Ravina. The Reporter. September, 2025.
- M5 The Burning Question: How can we live together with AI?⁹ William M. Mongan. Ursinus Magazine, October, 2023.
- M4 The New View: How an Ursinus lab is threading machine learning and the human element¹⁰. Ursinus Magazine, October, 2023.
- M3 How the Broad Street Run times thousands of runners at once¹¹. The Philadelphia Inquirer, April 26, 2022.
- M2 Bellyband and Exoskin Glove featured at Chemical Heritage Foundation, October, 2016.
- M1 Science Nation Coverage of Biomedical Smart Textiles: "These Smart Threads Could Save Lives"¹², September, 2016.

⁶<https://mainlinetoday.com/life-style/artificial-intelligence-degrees/>

⁷<https://philadelphia.today/2026/02/ursinus-college-s-stem/>

⁸<https://www.thereporteronline.com/2025/08/22/montgomery-county-appoints-members-to-ai-advisory-council/>

⁹https://www.ursinus.edu/live/news/8153-how-can-we-live-together-with-ai/_ingredients/templates/2021/magazine-story

¹⁰https://www.ursinus.edu/live/news/8149-the-new-view/_ingredients/templates/2021/magazine-story

¹¹<https://www.inquirer.com/health/broad-street-run-2022-races-in-philadelphia-20220426.html>

¹²https://nsf.gov/news/mmg/mmg_disp.jsp?med_id=132437&from=

HONORS,
CERTIFICATIONS,
AND PROFESSIONAL
MEMBERSHIP

HONORS AND AWARDS

- Colonial Academic Alliance Award for Sociology of Health and Wellness with Kelly Joyce, 2021-2022, \$4,000.
- Drexel College of Computing and Informatics Faculty Leadership Award, 2019 (Inaugural College Recipient)
- Drexel Center for the Advancement of STEM Teaching and Learning Excellence (CASTLE) Instructor of the Week, February, 2018
- Drexel College of Computing and Informatics (CCI) Teaching Excellence Award, 2014 (Inaugural Departmental Recipient)
- Drexel Dean's Fellow September, 2005 through September, 2008
- Member of the Drexel University Upsilon Pi Epsilon International Honor Society for the Computing Sciences since June, 2004; elected Treasurer August, 2006
- Member of the Golden Key International Honor Society January, 2001
- Member of the National Society of Collegiate Scholars January, 2001
- Upper Darby School District Service Award, 2000

PROFESSIONAL MEMBERSHIP

- Member of the National Council of Women in Information Technology (NCWIT), 2017-2019 (Drexel University), 2023-Present (Ursinus College)
- Member of the American Radio Relay League (ARRL), 2017-Present
- Member of the International Advisory Committee for the International Joint Journal Conference in Engineering (IJJCE) 2009
- Fellow of the Association of Computer Electronics and Electrical Engineers (ACEEE) since January, 2009
- Member of the Experimental Aircraft Association (EAA240) from May, 2008 through May, 2009
- Member of the National Council of Teachers of Mathematics (NCTM) from March, 2008 through March, 2009
- Student Member of the American Institute of Aeronautics and Astronautics (AIAA) from October, 2006 through September, 2008
- Senior Member of the Institute of Electrical and Electronics Engineers (IEEE) from 2005 through January, 2011, and since August, 2014 (elevated to Senior Member in October, 2018)
 - Member of the IEEE Computer Society
 - Member of the IEEE Signal Processing Society
 - Member of the IEEE Circuits and Systems Society
 - Member of the IEEE Information Theory Society
- Senior Member of the Association for Computing Machinery (ACM) since May, 2004 (elevated to Senior Member in March, 2012)
 - Member of the ACM Special Interest Group in Computer Science Education (SIGCSE)
 - Member of the ACM Computer Science Teachers Association (CSTA)
- Member of the Aircraft Owners and Pilots Association (AOPA) since May, 2004

PROFESSIONAL, SAFETY, AND MISCELLANEOUS CERTIFICATIONS

- FCC GMRS License with callsign WSKN786, 2025
- Volunteer Examiner (VE) for FCC Amateur Radio Licensure under the ARRL Volunteer Examiner Coordinators (VEC)
- Member of the Montgomery County Amateur Radio (MCAR) Emergency Service (ARES) / Radio Amateur Civil Emergency Service (RACES) Unit, 2024-Present
 - SKYWARN[®] Net Control for severe weather activations and monthly practice nets in South-eastern Pennsylvania
 - Net Control for MCAR ARES/RACES Mesh Chat, Voicemail, and Conference net
- ARES Basic and Intermediate Training, 2024
 - ARRL Basic EmComm Course
 - ARRL Intermediate EnComm Course
 - National Weather Service (NWS) SKYWARN[®] - Spotter Basic Training
 - FEMA IS-100: Intro to Incident Command System
 - FEMA IS-200: ICS for Single Resource
 - FEMA IS-700: Introduction to National Incident Mgt. System

- FEMA IS-800: National Response Framework
- Wine Road Ambassador, 2024
- FAA Unmanned Aircraft General Certification for Part 107 Commercial Drone Operations, 2023
 - FAA Unmanned Aerial Vehicle (UAV) Certificate FA3TN3LPL4, 2017
- Vermont State Bartending License (Alcohol Safety Course), 2019
- Pennsylvania Boater Safety Certification, 2017
- FCC Amateur Radio License with callsign W1CLK (Technician: 2017, General: 2017, Amateur Extra: 2019)
- FCC Restricted Radiotelephone Operator (RR) License, 2016
- Bioraft Laboratory Safety Certification 2016-2025
 - Electrical Safety
 - Emergency Response
 - Fire and Life Safety
 - Hazard Communication
 - Hazardous Waste Management
 - Personal Protection Equipment
 - Respiratory Protection
- Human Subjects Protection Certification 2015-Present
 - Medical Good Clinical Practice
 - Medical Responsible Conduct of Research
 - Social and Behavioral Conflict of Interest
 - Medical Human Subjects Research
 - Social and Behavioral Human Subjects Research
 - Health Information Privacy and Security - Medical, Biomedical, Nursing, Public Health, and Psychology
 - HIPAA and Medical Research
 - Responsible Conduct of Research for Engineers
 - Social and Behavioral Responsible Conduct of Research
- Pennsylvania Instructional I Teaching Certificate earned September, 2008
 - Grades 7-12 Mathematics Subject Area earned January, 2007
 - Grades 7-12 Computer Science Subject Area earned July, 2021
- United States Department of Defense Antiterrorism Level 1 Awareness Training
- Security clearance at the level of SECRET granted April, 2006
- FAA Certificated Private Pilot, 2005-Present
 - Instrument Rating (Instrument Airplane) March, 2006
 - Complex Endorsement (Airplane Single Engine Land), September, 2005
 - Private Pilot (Airplane Single Engine Land) May, 2005

SELECTED
PRE-FACULTY
PROFESSIONAL
EXPERIENCE

Research Assistant Drexel University
Philadelphia, Pennsylvania 2001-2008

- Researched projects and tools for reverse engineering, software maintenance and program understanding with the Software Engineering Research Group (SERG)
- Developed REportal: A Web Based Reverse Engineering Portal and re-engineered it into a maintainable, distributed Service Oriented Architecture
- Developed Abstract Software Cluster Visualization tool using XML

Technical Writer and Research Assistant ACIN Program at Drexel University
Camden, New Jersey 2005-2008

- Performed research, development and technical writing related to reverse engineering and software architecture with US Army CERDEC RDECOM C2D under the Applied Communications and Information Networking (ACIN) Project
- Served as a contributing member of the US Army Intelligent Agent (IA) sub-IPT group
- Researched tools and best practices to investigate software engineering and reverse engineering methods for distributed, multi-agent systems
- Co-authored the Agent Systems Reference Model (ASRM) and the Agent Systems Reference

Architecture (ASRA)

Application Developer and IT Manager David M. Banet and Associates
 Exton, Pennsylvania 2001-2005

- Developed customized software solutions to simplify business tasks, including:
 - Payroll integration system to compute payroll benefit deductions and integrate with client proprietary payroll systems
 - XML based electronic audit system
 - Specialized web-based services for clients
- Built and maintained network domain and all associated servers, workstations and integrated business tools

IT Consultant Consulting Group, Inc.
 Malvern, Pennsylvania 2004

- Served as IT technician for the office during Open Enrollment period
- Assisted in the maintenance of network domain and associated workstations
- Provided technical support in troubleshooting workstations and servers

Volunteer Chester-Wallingford Red Cross
 Chester, Pennsylvania 2003

- Replaced token-ring computer network with 10/100 ethernet and internet connectivity on a volunteer basis

Instructional Technology Consultant Upper Darby School District
 Upper Darby, Pennsylvania 1998-2003

- Designed and managed web site for Upper Darby School District
- Created Upper Darby School District Instructional Intranet (RoyalNet) and developed systems to automate content management
- Presented Act 48 training workshops on Filemaker Pro and Dreamweaver for staff, teachers and principals, to maintain and add value to the production of the above district-wide systems
- Interviewed for an Upper Darby School Board seat in November, 2000
- Served on the Upper Darby Technology and Grant Writing Committee under the school board in March, 2001

Application Developer and Consultant StarComm Development, Inc.
 Upper Darby, Pennsylvania 1999-2002

- Developed and maintained dynamic, database-driven internet applications (ASPs) for clients
- Provided development support on FileMaker programming and educational projects for various companies and educational institutions
- Assisted in the development process and provided technical support for Drexel University's online college application system

Web Application Developer Boss Entertainment
 King of Prussia, Pennsylvania 2001

- Designed, developed, managed and maintained e-Commerce web site
- Supported development of golf course management system

Script Reviewer Bergwall Productions
 Kennett Square, Pennsylvania 1998-1999

- Read, reviewed, and edited scripts for Bergwall Productions instructional computer videos